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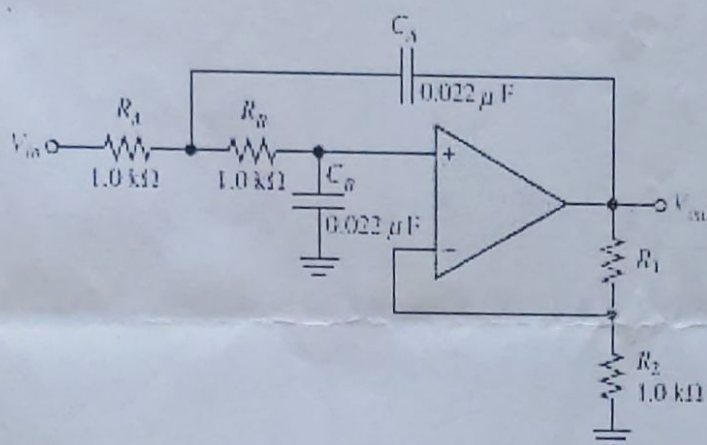
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MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY (COICT)
ET 8213 TEST TWO 2023/2024.

Time: 60min

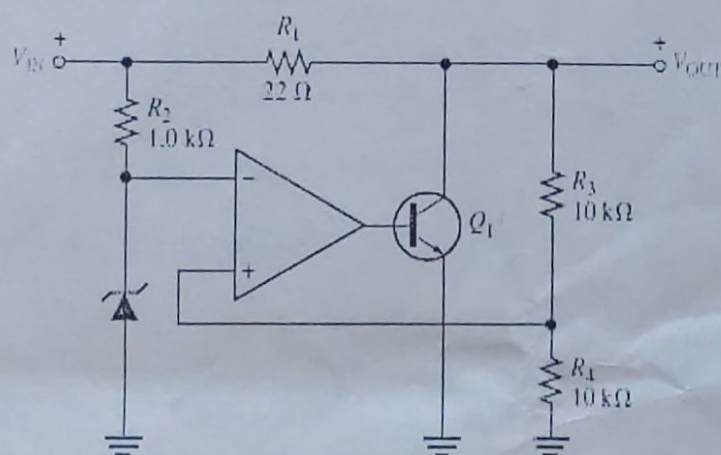
QUESTION ONE [10 Marks]

- Explain the concept of cutoff frequency in the context of electronic filters.
- Describe four types of active Filters.
 - Draw the corresponding response curve for each of the filter above.
 - Write the practical application for each of the filter above.
- Give two basic difference between active filter and passive filter.
 - Determine the critical frequency of the Sallen-Key low-pass filter in figure below, and set the value of R_1 for an approximate Butterworth response.



QUESTION TWO [5 Marks]

- Explain the difference between a series linear regulator and a shunt linear regulator.
- What power rating must R_1 have if the maximum input voltage is 12.5 V in the circuit below?



$$P = \frac{V^2}{R}$$

12V 1W 10kΩ
 12V 1W 10kΩ

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BONIFACE 'B' SANKWA

1. (a). Cutoff frequency is the frequency whose output is 70.7 percent of input. It is used to determine performance characteristics of filter. Also cutoff frequency are boundary between bandpass and bandstop.

(b) (i). Types of active filters

1. ~~to~~ Active low filter. This allow only low frequency ~~to~~ and reject others.

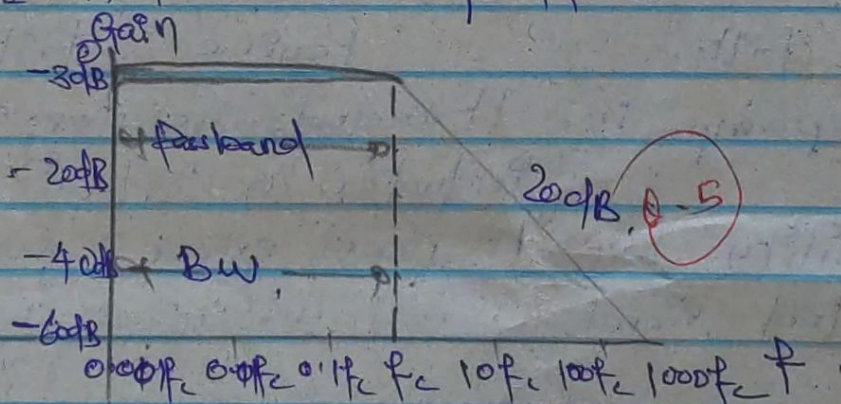
2. Active high filter. This allow only high frequency and reject others.

3. ~~B~~ Active band pass filter. This allow a certain band of frequency.

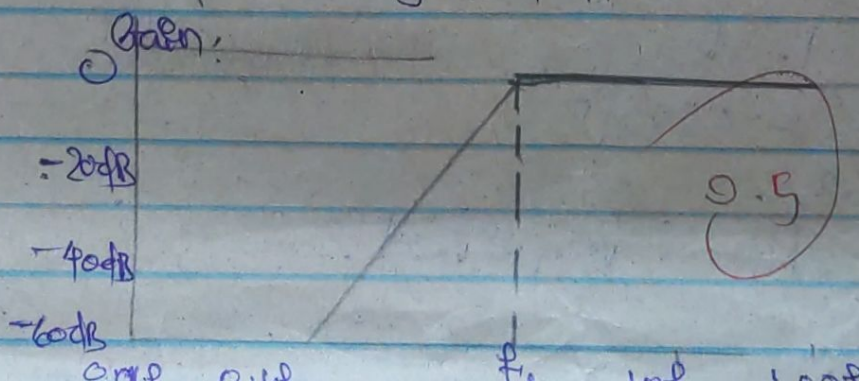
4. Active bandstop filter. This attenuate certain band of frequency.

(ii). Response curve.

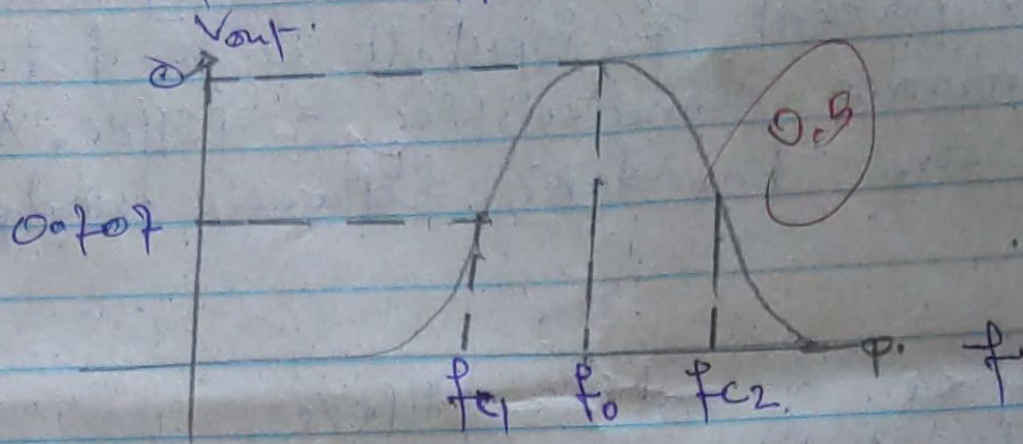
1. Active low filter.



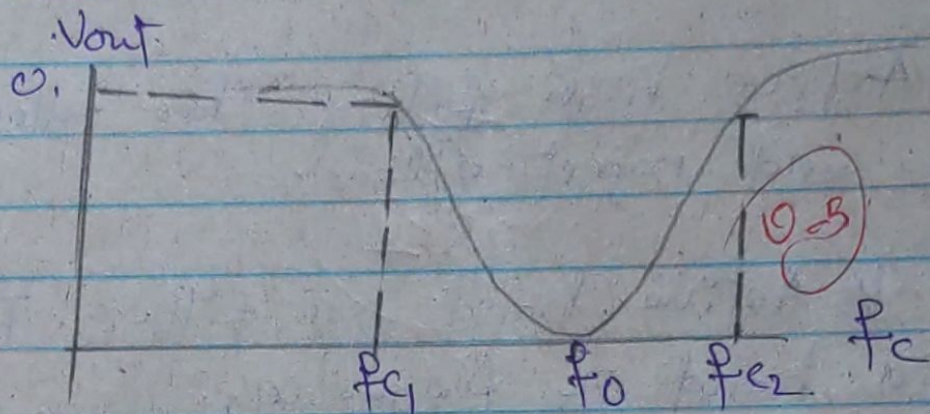
2. Active high filter.



1. (b) (ii) 3. Active bandpass filter.



4. Active bandstop filter.



(iv) Application of filter.

1. Signal processing.
2. Communication system.
3. Audio system.
4. Noise reduction.

(c) (i). Active filter

1. It includes opamp in circuit.
2. It has the ability to reject unwanted signal itself.

Passive filter.

- It includes resistor, capacitor, inductor in circuit.
- It needs an external source to reject unwanted signal.

(ii). Given

$$R_A = R_B = 1.0 \text{ k}\Omega$$

$$C_A = C_B = 0.022 \mu\text{F}$$

$$R_2 = 1.0 \text{ k}\Omega$$

f_c and R_1 required.

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(e) (i) from, $f_c = \frac{1}{2\pi RC}$

$$f_c = \frac{1}{2\pi \times 1k\Omega \times 0.022\mu F}$$

then, $f_c = 7.23 \text{ KHz}$

$$\frac{R_1}{R_2} = 0.586$$

$$R_1 = 0.586 R_2$$

$$R_1 = 0.586 \times 1.0k\Omega$$

$$R_1 = 0.586 k\Omega$$

The critical frequency is 7.23 KHz and $R_1 = 0.586 k\Omega$

2. (a) Series linear regulator | shunt linear regulator.
 (i) It is more efficiency | It is less efficiency.

(ii) It able to produce more output voltage than its input. | It can't produce more output voltage than its input.

(b) Given

$$V_{in} = 12.5 \text{ V}$$

$$R_1 = 22\Omega$$

Power

$$P = V_{in} \times I_{R_1}$$

$$P = 12.5 \text{ V} \times 22\Omega$$

$$P = 275 \text{ W}$$

$$P = IV = \frac{V^2}{R}$$

$$P = \frac{(12.5)^2}{22}$$

$$P = 7.102 \text{ W}$$

The power at R_1 is 7.102 W